

Problem Set 1

Preferences and Utility Functions

EC306: Intermediate Microeconomics

Instructions. Work through these in class. Show all working and interpret your numerical answers in words. Worked solutions follow at the end.

Question 1 — Axioms and their limits

- (a) State the completeness and transitivity axioms formally.
- (b) A consumer ranks three holidays: Lisbon $\succ$ Athens, Athens $\succ$ Rome, Rome $\succ$ Lisbon. Construct an explicit “money-pump” sequence of three trades, each improving the consumer’s bundle by their own ranking, that returns them to the start strictly poorer. State the total loss if each trade costs $\pounds c$.
- (c) In lecture we argued the axioms are a modelling device, not literal truths. Give *one* concrete, named real-world phenomenon (e.g. a documented behavioural effect) that violates transitivity, and explain in 3–4 sentences why economists nonetheless retain the axiom. Generic answers (“people are irrational”) score zero; name the effect and tie it to the axiom.

Question 2 — MRS, convexity, and a general CES

Consider the CES utility $u(x, y) = (\alpha x^\rho + (1 - \alpha)y^\rho)^{1/\rho}$, with $0 < \alpha < 1$ and $\rho < 1$, $\rho \neq 0$.

- (a) Derive MU_x , MU_y , and show that $MRS = \frac{\alpha}{1 - \alpha} \left(\frac{x}{y}\right)^{\rho-1}$.
- (b) Show that as $\rho \rightarrow 1$ the MRS becomes independent of (x, y) , and interpret what kind of goods this limiting case describes.
- (c) Take the limit $\rho \rightarrow 0$. Show the preferences converge to Cobb–Douglas and identify the resulting exponents. (Hint: examine $\ln u$ and apply l’Hôpital in ρ .)
- (d) For $\rho = -1$, $\alpha = 0.5$, evaluate the MRS at $(x, y) = (1, 4)$ and at $(4, 1)$. In two sentences, explain what the *change* in the MRS between these points tells you about how substitutable the two goods are here, relative to the Cobb–Douglas case.

Question 3 — Representation and transformations

- (a) Define precisely what it means for $u(\cdot)$ to *represent* a preference relation.
- (b) You are given two utility functions:

$$u_1(x, y) = x^{1/3}y^{2/3}, \quad u_2(x, y) = \ln x + 2 \ln y.$$

Determine whether they represent the same preferences. If so, exhibit the exact monotonic transformation g such that $u_2 = g(u_1)$; if not, prove they differ by finding a pair of bundles ranked oppositely.

- (c) Now consider $u_3(x, y) = x^{1/3}y^{2/3} + 0.01x$. Is u_3 a monotonic transformation of u_1 ? Justify carefully — compute and compare the two MRS expressions, and explain what economically changes when the extra term is added.

Question 4 — Functional forms meet the real market

This question requires **current data you locate yourself**. Cite each source with a URL and the date you accessed it.

- (a) Find the *current* retail price of one specific inkjet printer model and a single compatible OEM ink cartridge for it. Report both prices, the model, and your sources.
- (b) Compute the ratio (annualised cartridge spend for a stated, justified usage level) to (printer price). State your usage assumption explicitly.
- (c) Using the perfect-complements model from lecture, explain why the observed pricing pattern (cheap hardware, expensive consumable) is consistent with $u = \min\{x, y\}$ -type demand. Be specific about which good carries the margin and why the fixed-ratio property enables this.
- (d) Identify *one* feature of the real market that the pure perfect-complements model misses (e.g. third-party cartridges, refilling, chipped cartridges). Explain in 3–4 sentences how that feature would change the consumer's effective MRS and therefore the firm's ability to extract margin.

Solutions

Question 1

(a) *Completeness*: for any two bundles a, b , either $a \succeq b$ or $b \succeq a$ (every pair can be ranked).
Transitivity: for any a, b, c , if $a \succeq b$ and $b \succeq c$ then $a \succeq c$.

(b) The ranking cycles: $L \succ A$, $A \succ R$, $R \succ L$. Start holding Athens. (1) $L \succ A$: pay £ c to swap $A \rightarrow L$. (2) $R \succ L$: pay £ c to swap $L \rightarrow R$. (3) $A \succ R$: pay £ c to swap $R \rightarrow A$. The consumer is back at Athens having paid £ $3c$. Total loss = **3c**.

(c) *Model answer*. A named violation: Tversky's (1969) experimental *intransitivity of preferences* (a lexicographic-semiorder pattern over multi-attribute gambles), or documented preference-reversal/framing effects. Economists keep the axiom because (i) the violations are systematic and typically small in market choices; (ii) without transitivity there is no well-defined "most preferred" bundle, so demand is not pinned down and comparative statics vanish; (iii) it is used as an "as-if"/normative benchmark whose predictions fit aggregate data well. "People are irrational" alone is not an answer.

Question 2

(a) Writing $S = \alpha x^\rho + (1 - \alpha)y^\rho$, $u = S^{1/\rho}$:

$$MU_x = \frac{1}{\rho} S^{1/\rho-1} \cdot \alpha \rho x^{\rho-1} = S^{1/\rho-1} \alpha x^{\rho-1}, \quad MU_y = S^{1/\rho-1} (1 - \alpha) y^{\rho-1}.$$

$$MRS = \frac{MU_x}{MU_y} = \frac{\alpha x^{\rho-1}}{(1 - \alpha) y^{\rho-1}} = \frac{\alpha}{1 - \alpha} \left(\frac{x}{y} \right)^{\rho-1}.$$

(b) As $\rho \rightarrow 1$, $(x/y)^{\rho-1} \rightarrow (x/y)^0 = 1$, so $MRS \rightarrow \alpha/(1 - \alpha)$, a constant independent of (x, y) . Constant MRS means **linear indifference curves**: the goods are *perfect substitutes*.

(c) Let $\ln u = \frac{1}{\rho} \ln(\alpha x^\rho + (1 - \alpha)y^\rho)$. As $\rho \rightarrow 0$ this is $0/0$. By l'Hôpital in ρ ,

$$\lim_{\rho \rightarrow 0} \ln u = \frac{\alpha x^\rho \ln x + (1 - \alpha)y^\rho \ln y}{\alpha x^\rho + (1 - \alpha)y^\rho} \bigg|_{\rho=0} = \alpha \ln x + (1 - \alpha) \ln y,$$

so $u \rightarrow x^\alpha y^{1-\alpha}$: **Cobb–Douglas** with exponents α and $1 - \alpha$.

(d) With $\rho = -1$, $\alpha = 0.5$: $MRS = (x/y)^{-2} = (y/x)^2$. At $(1, 4)$: $MRS = 16$; at $(4, 1)$: $MRS = 1/16$. The MRS swings by a factor of 256 across these points, versus a factor of 16 for Cobb–Douglas ($MRS = y/x$ gives 4 and $1/4$). The steeper swing means the two goods are *less* substitutable here ($\sigma = 1/(1 - \rho) = \frac{1}{2} < 1$): the marginal valuation reacts much more sharply to the bundle mix.

Question 3

(a) u represents $\succeq$ if for all bundles a, b : $u(a) \geq u(b) \iff a \succeq b$.

(b) $\ln u_1 = \frac{1}{3} \ln x + \frac{2}{3} \ln y$, so $3 \ln u_1 = \ln x + 2 \ln y = u_2$. Hence $u_2 = g(u_1)$ with $g(t) = 3 \ln t$, strictly increasing for $t > 0$. They represent **the same preferences**.

(c) $MRS_1 = \frac{(1/3)x^{-2/3}y^{2/3}}{(2/3)x^{1/3}y^{-1/3}} = \frac{y}{2x}$. For u_3 , $MRS_3 = \frac{(1/3)x^{-2/3}y^{2/3} + 0.01}{(2/3)x^{1/3}y^{-1/3}} \neq \frac{y}{2x}$. A monotonic transformation leaves the MRS (hence every indifference curve) unchanged; here the MRS differs, so u_3 is **not** a monotonic transform of u_1 — it is a different preference. Economically the extra $0.01x$ adds value to x that does not depend on y , raising the marginal valuation of x and tilting the indifference curves.

Question 4 (model answer / marking notes)

Data-dependent; a strong answer: **(a)** reports a current, cited printer price and a current OEM cartridge price (URL + access date). **(b)** states an explicit usage assumption (e.g. pages/month $\Rightarrow$ cartridges/year) and computes annual cartridge spend $\div$ printer price; the ratio is typically well above 1. **(c)** maps to $u = \min\{\text{printer}, \text{ink}\}$: with fixed-proportions demand the consumer is locked into buying ink once the printer is owned, so the firm puts the margin on the *consumable* (ink), prices the hardware near or below cost, and harvests the captive aftermarket. **(d)** a real feature the model misses — third-party/refilled/chipped cartridges — makes ink a partial *substitute* rather than a pure complement, flattening the effective MRS and eroding the firm's ability to extract aftermarket margin (hence DRM chips).